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Optimization of In-Situ Bio-Surfactant Production in Microbial Improved Oil Recovery

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Abstract

Microbial Improved Oil Recovery (MIOR) has the potential to be one of the most cost-effective IOR methods. However, uncertainties about different MIOR mechanisms and their immediate returns on investment (ROI) have prevented their wider adoption. Here, we demonstrate that applying state-of-the-art optimization algorithms to the in-situ bio-surfactant production process can result in outstanding financial benefits. Furthermore, sensitivity studies show consistently high ROI with varying microbial conditions and costs for a synthetic reservoir model.

We performed numerical experiments on a reservoir simulator with well-calibrated in-situ bio-surfactant production functionality. Different costs for MIOR (including costs of microbes and nutrients) as well as oil prices were considered in calculating the net present values (NPV) for various MIOR processes. In addition, a novel Ensemble Optimization (EnOpt) algorithm (originally developed to optimize injection and production rates in waterflooding) was used to adjust microbe and nutrient injection concentrations dynamically during the MIOR processes to further maximize NPV.

We compared the final NPV's after (1) only waterflooding, (2) MIOR without optimization, and (3) MIOR with optimization. The in-situ bio-surfactant production showed remarkable 5- to 10-fold NPV increases for MIOR without and with optimization, respectively, compared to only waterflooding. In addition, the cumulative oil production increased with an extra 15 to 20% of original oil in place (OOIP) for the synthetic reservoir model after MIOR. Finally, sensitivity studies showed that the NPV increases were consistently above 10-fold for a wide range of microbe prices, indigenous reservoir microbe concentrations, and oil prices for MIOR with optimization compared to only waterflooding.

To the best of our knowledge, this paper is the first study that quantifies the ROI of MIOR measured in NPV with realistic assumptions of the costs and rewards in an exemplary in-situ bio-surfactant production process. The observed high NPV increases were promising and encouraging for further developments in this direction. Finally, successful application of the EnOpt algorithm to the MIOR problem has not been reported in literature and is worth highlighting here.

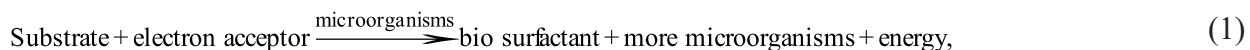
Introduction

Microbial Improved Oil Recovery (MIOR) has emerged as a promising technique in the oil and gas industry, leveraging the capabilities of microorganisms to enhance the extraction of hydrocarbons from reservoirs. This method, which involves the injection of specific microbial strains into oil wells, aims to improve oil recovery rates by altering the physical and chemical properties of the reservoir. Recent advancements in MIOR have demonstrated significant potential, driving innovation and sustainability (Patel et al. 2015, Niu et al. 2020). Studies have shown that microbial activity can lead to the production of biosurfactants, biopolymers, and gases, which collectively enhance oil mobility and reduce viscosity. These microbial processes can also modify reservoir permeability and porosity, facilitating the displacement of trapped oil. The integration of MIOR into existing recovery methods has resulted in notable improvements in oil yield, for example in reference (Xiaolin et al. 2012). Academically, the exploration of MIOR has expanded significantly, with researchers focusing on the identification and optimization of microbial strains, understanding microbial interactions within reservoirs, and developing predictive models for MIOR performance (Ziwei et al. 2025). Advances in molecular biology and biotechnology have enabled the characterization of microbial communities and their metabolic pathways, providing insights into the mechanisms underlying MIOR. Collaborative efforts between academia and industry have led to the development of innovative MIOR technologies, fostering a deeper understanding of microbial ecology and its application in oil recovery.

Even though MIOR has been developed for decades, uncertainties about different MIOR mechanisms and their immediate returns on investment (ROI) have prevented their wider adoption. To this end, the present study aims at quantifying the ROI by calculating the net present values (NPV) for various MIOR processes with well-calibrated reservoir simulator. Furthermore, novel production optimization algorithm (originally developed to optimize injection and production rates in waterflooding) is used to adjust microbe and nutrient injection concentrations dynamically during the MIOR processes to further maximize NPV.

Methods

In this study, we used a well-calibrated reservoir simulator (UTCHEM) and its detailed input parameters with the in-situ bio-surfactant production functionality. The chemical fluid and surfactant phase behavior model parameters, biodegradation kinetic parameters, and relative permeability parameters for the water, oil, and microemulsion phases were adopted from (Hosseini-noosheri, Lashgari, and Sepehrnoori 2016). Biodegradation involves oxidation-reduction reactions facilitated by microorganisms. This process occurs through the transfer of electrons from an electron donor (substrate) to an electron acceptor in the presence of these microorganisms. This study uses the generation of bio-surfactants through the biodegradation process as outlined below:



The biological reaction model in UTCHEM enhances (Molz, Widdowson, and Benefield 1986) model to accommodate an unlimited number of substrates, electron acceptors, and biological species. This model accounts for microorganisms that are either attached or free-floating. Detailed information on the assumptions and capabilities of this model is provided by (De Blanc, McKinney, and Speitel Jr 1996). The basic equations for a single substrate, electron acceptor, and biological species are the following:

$$\frac{dS}{dt} = -\frac{\beta k \bar{X}}{m_c} (S - \bar{S}) - \frac{\mu_{\max} X}{Y} \left(\frac{S}{K_s + S} \right) \left(\frac{A}{K_A + A} \right) - k_{\text{abio}} S, \quad (2)$$

$$\frac{d\bar{S}}{dt} = \frac{\beta k}{V_c} (S - \bar{S}) - \frac{\mu_{\max} \rho X}{Y} \left(\frac{\bar{S}}{K_s + \bar{S}} \right) \left(\frac{\bar{A}}{K_A + \bar{A}} \right) - K_{\text{abio}} \bar{S}, \quad (3)$$

$$\frac{dA}{dt} = -\frac{\beta k \bar{X}}{m_c} (A - \bar{A}) - \frac{\mu_{\max} X E}{Y} \left(\frac{S}{K_s + S} \right) \left(\frac{A}{K_A + A} \right), \quad (4)$$

$$\frac{d\bar{A}}{dt} = \frac{\beta k}{V_c} (A - \bar{A}) - \frac{\mu_{\max} \rho_X E}{Y} \left(\frac{\bar{S}}{K_s + \bar{S}} \right) \left(\frac{\bar{A}}{K_A + \bar{A}} \right), \quad (5)$$

$$\frac{dX}{dt} = \mu_{\max} X \left(\frac{s}{K_s + S} \right) \left(\frac{A}{K_A + A} \right) - bX, \quad (6)$$

$$\frac{d\bar{X}}{dt} = \mu_{\max} \bar{X} \left(\frac{\bar{S}}{K_s + \bar{S}} \right) \left(\frac{\bar{A}}{K_A + \bar{A}} \right) - b\bar{X}, \quad (7)$$

where

S = aqueous phase substrate concentration (ML^{-3})

$\bar{S}$ = substrate concentration in attached biomass (ML^{-3})

A = aqueous phase electron acceptor concentration (ML^{-3})

$\bar{A}$ = electron acceptor concentration in attached biomass (ML^{-3})

X = aqueous phase (unattached) biomass concentration (ML^{-3})

$\bar{X}$ = attached biomass concentration; mass of attached cells per volume of aqueous phase (ML^{-3})

β = surface area of a single microcolony (L^2)

k = mass transfer coefficient (LT^{-1})

$\mu_{\max}$ = maximum specific growth rate (T^{-1})

m_c = mass of cells in a single microcolony

E = mass of electron acceptor consumed per mass of substrate biodegraded

ρ_X = biomass density; mass of cells per volume of biomass (ML^{-3})

V_c = volume of a single microcolony (L^3)

Y = yield coefficient; mass of cells produced per mass of substrate biodegraded

K_s = substrate half - saturation coefficient (ML^{-3})

K_A = electron acceptor half - saturation coefficient (ML^{-3})

k_{abio} = first - order reaction rate coefficient (for abiotic decay reactions, T^{-1})

b = endogenous decay coefficient (T^{-1})

t = time (T)

Eqs. (2) through (7) cover mechanisms for substrate and electron acceptor loss, as well as the growth rates of attached and unattached biomass. Eqs. (2) and (3) detail three mechanisms for substrate loss: (a) First term: diffusion of substrate from bulk liquid across a stagnate liquid film; (b) Second term: biodegradation of substrate in bulk liquid (Monod's expression); (c) Third term: abiotic loss of substrate through first-order reactions. Equations (4) and (5) demonstrate the loss of electron acceptors through a biodegradation reaction that accounts for both diffusion and Monod's expression as distinct components. Equations (6) and (7) detail the growth and decay processes of unattached and attached biomass, respectively. In this study, we define the reaction where a substrate (glucose) interacts with an electron acceptor (sulfate) in the presence of microbes (*Bacillus*) to produce surfactant. Table 1 outlines the parameters involved in the injection of microbes and nutrients, as well as the reactions that facilitate the production of bio-surfactants:

Table 1—Biodegradation kinetic parameters used in the simulation.

Parameters	
Initial substrate concentration, S_0 (mg/L)	0
Initial electron acceptor concentration, A_0 (mg/L)	0
Initial biomass concentration, X_0 (mg/L)	5.0
Biomass maximum specific growth rate, $\mu_{\max}$ (d ⁻¹)	3.0
Biomass endogenous decay coefficient, b (d ⁻¹)	0.18
Yield coefficient, Y (mass X/mass S)	1.0
Electron acceptor utilization coefficient, E (mass A/mass S)	0.35
Initial biomass concentration, X (mg/L)	5.0
Half-saturation coefficient for substrate, K_s (mg/L)	7.0
Half-saturation coefficient for electron acceptor, K_A (mg/L)	0.77

Water flooding often leaves behind residual oil due to the interfacial tension between oil and water. The in-situ biosurfactant interacts with oil and water at various salinities, creating three types of microemulsions: Type II (-) at low salinity, Type II (+) at high salinity, and Type III at optimal salinity. To effectively remove the residual oil, the biosurfactant and microemulsion must lower the interfacial tension, allowing both oil and water to flow continuously. Moreover, the biosurfactant must maintain low interfacial tension at the displacement front to prevent the re-trapping of mobilized oil by capillary forces, which requires concentrations above the Critical Micelle Concentration. (Youssef et al. 2007) demonstrated that in-situ biosurfactants are sufficient to mobilize trapped oils.

To model the IFT reduction in the MIOR process, biosurfactant must be generated within a biological model. The existing IFT model in the simulator will then reduce the interfacial tension (IFT) between oil and water. This model adopts a relative permeability alteration approach to simulate IFT reduction between oil and water. Within the simulator, relative permeability varies according to different trapping numbers. When biosurfactant is generated in a gridblock, it decreases the IFT, thereby increasing the trapping number. The reduction of IFT and the mobilization of oil result in a decrease in residual oil saturation, which subsequently alters the endpoint relative permeabilities and exponents. For the governing equations for IFT reduction in surfactant flooding, we refer the readers to the UTCHEM Technical Document (Delshad, Pope, and Sepehrnoori 2000). The bio-surfactant model parameters are summarized in Table 2:

Table 2—Bio-surfactant model parameters.

Parameters	
Microemulsion viscosity parameters	$a_1 = 0 \text{ wt}\%^{-1}$, $a_2 = 0 \text{ wt}\%^{-1}$, $a_3 = 1 \text{ wt}\%^{-1}$, $a_4 = 0 \text{ wt}\%^{-1}$, $a_5 = 1.7 \text{ wt}\%^{-1}$
CMC	0.0005 vol/vol
Optimal salinity	0.598 meq/ml (32,000 ppm TDS)
Lower optimal salinity	0.45 meq/ml (44,000 ppm TDS)
Upper optimal salinity	0.9 meq/ml (27,000 ppm TDS)
Height of binodal curve	$H_0 = 0.0017$, $H_1 = 0.0017$, $H_2 = 0.0017$
Bio-surfactant adsorption parameters	$A_{D31} = 1.1$, $A_{D32} = 0.21 \text{ ml/meq}$, $B_{D3} = 1000 \text{ vol/vol}$

The endpoints and exponents in relative permeability functions are modified due to the generation of bio-surfactants. The model calculates this modification using linear interpolation between the specified input values at low and high trapping numbers:

$$k_{rl}^o = k_{rl}^{olow} + \frac{s_{lr}^{low} - s_{lr}^{high}}{s_{lr}^{low} - s_{lr}^{high}} (k_{rl}^{ohigh} - k_{rl}^{olow}), \quad (8)$$

$$n_l = n_l^{low} + \frac{s_{lr}^{low} - s_{lr}^{high}}{s_{lr}^{low} - s_{lr}^{high}} (n_l^{high} - n_l^{low}), \quad (9)$$

The relative permeability parameters are summarized in Table 3:

Table 3—Relative permeability model parameters.

Phase	Parameters					
Water	$s_{rw}^{high} = 0$	$s_{rw}^{low} = 0.23$	$n_w^{high} = 1$	$n_w^{low} = 3.182$	$K_{rw}^{ohigh} = 1$	$K_{rw}^{olow} = 0.138$
Oil	$s_{ro}^{high} = 0$	$s_{ro}^{low} = 0.39$	$n_o^{high} = 1$	$n_o^{low} = 2.405$	$K_{ro}^{ohigh} = 1$	$K_{ro}^{olow} = 0.515$
Microemulsion	$s_{rm}^{high} = 0$	$s_{rm}^{low} = 0.23$	$n_m^{high} = 1$	$n_m^{low} = 3.182$	$K_{rm}^{ohigh} = 1$	$K_{rm}^{olow} = 0.138$

The synthetic reservoir model used here consisted of $9 \times 9 \times 40$ grid blocks of size 25 ft $\times$ 25 ft $\times$ 3 ft with total reservoir size of 225 ft $\times$ 225 ft $\times$ 120 ft. The porosity was set to 24 %, and the heterogeneous permeabilities were set between 10 to 500 mD across the vertical directions. The water density, oil density, water compressibility, and rock compressibility were set to 62.9 lb/ft³, 49.8 lb/ft³, 3×10^{-6} psi⁻¹, and 5.2×10^{-6} psi⁻¹, respectively. Initial pressure, temperature, and water saturation were set to 5600 psi, 160 °F, and 0.23, respectively. One injector and producer pair were placed at the opposite corners of the reservoir model. Pure water flooding was performed for the first 1000 days followed by microbe injection for another 1000 days with injection rate of 3000 ft³/day. Both constant and dynamic optimization microbe injections were used to compare their oil recovery performances.

The Ensemble Optimization (EnOpt) originally developed to optimize injection and production rates in waterflooding (Ow et al. 2018, Chen and Poitzsch 2018, Chen, Ow, and Poitzsch 2021) were adopted to optimize microbe injections. Given a scalar objective function $J(\mathbf{u})$, where $\mathbf{u}$ is a N_u -dimensional column vector including the control parameters, the EnOpt implementation is given by

$$\mathbf{u}^{l+1} = \mathbf{u}^l + \alpha_l \frac{\mathbf{C}_U \mathbf{g}_l}{\|\mathbf{C}_U \mathbf{g}_l\|_\infty}, \quad (10)$$

for $l = 0, 1, \dots$ until convergence. α_l is a step size, $\mathbf{C}_U$ is a predefined smoothing covariance matrix, and $\mathbf{g}_l$ is an approximate gradient column vector calculated by

$$\mathbf{g}_l = \frac{1}{N_e} \sum_{j=1}^{N_e} (\hat{\mathbf{u}}_j - \mathbf{u}^l) [J(\hat{\mathbf{u}}_j) - J(\mathbf{u}^l)], \quad (11)$$

with N_e denoting the number of ensemble members, and $\hat{\mathbf{u}}_j$ is the perturbed control parameters according to $\hat{\mathbf{u}}_j \sim N(\mathbf{u}_p, \mathbf{C}_U)$. In the microbe injection optimization. The control parameters and objective function are

$$\mathbf{u} = [c_{Bacillus}^1, c_{glucose}^1, c_{sulfate}^1, \dots, c_{Bacillus}^N, c_{glucose}^N, c_{sulfate}^N]^T, \quad (12)$$

$$J(\mathbf{u}) = NPV = NPV_{oil} - MIOR \text{ Cost} = \frac{\sum_{n=1}^N r_o q_o^n \Delta t^n}{(1+b)^{n/365}} - MIOR \text{ Cost}, \quad (13)$$

with $c^n_{component}$ is the injected concentration of each MIOR component at the n^{th} step; r_o is the oil price; q_o is the oil production rate; b is a discount rate; N is the total optimization steps with t^n denoting the elapsed time in days and Δt^n denoting the step size of each time step. In this study, we used $\mathbf{C}_U = \mathbf{I}$, $N_e = 240$, $N = 20$, $b = 0.1$, and $\Delta t^n = 50$ days for the EnOpt algorithm with NPV calculations. Different oil prices and MIOR costs were used for sensitivity analysis.

The convergence criteria were defined as follows (Do and Reynolds 2013): (i) In each iteration l after calculating the approximate gradient $\mathbf{g}_l$, an initial step size of $\alpha_l = 1$ was used to calculate $\mathbf{u}^{l+1}$. If the new control parameters did not increase NPV, i.e., $\frac{1}{N_e} \sum_{j=1}^{N_e} J_f(\mathbf{u}^{l+1}) < \frac{1}{N_e} \sum_{j=1}^{N_e} J_f(\mathbf{u}^l)$, α_l was cut in half and $\mathbf{u}^{l+1}$ was recomputed with this reduced step size. If after five step size cuts the new control parameters still did not increase NPV, then we generated a new set of perturbed controls $\hat{\mathbf{u}}_j$ and calculated a new $\mathbf{g}_l$. If an increase of NPV was not found after five successive resampling of the perturbations, then the algorithm was terminated. (ii) The algorithm was also terminated if the relative increase in the objective function was less than 10^{-4} , i.e., $\left| \frac{1}{N_e} \sum_{j=1}^{N_e} J_f(\mathbf{u}^{l+1}) - \frac{1}{N_e} \sum_{j=1}^{N_e} J_f(\mathbf{u}^l) \right| \left| \frac{1}{N_e} \sum_{j=1}^{N_e} J_f(\mathbf{u}^l) \right| \leq 10^{-4}$, and the l_2 norm of the relative change in the control vector was less than 10^{-3} , i.e., $\|\mathbf{u}^{l+1} - \mathbf{u}^l\|_2 / \max(\|\mathbf{u}^l\|_2, 1) \leq 10^{-3}$.

Results and Discussion

In the baseline scenario, we used the oil price of $r_o = \$70/\text{bbl}$ with the MIOR costs for *Bacillus*, glucose, and sulfate of $\$30/\text{kg}$, $\$400/\text{ton}$, and $\$150/\text{ton}$, respectively. Fig. 1 shows the constant as well as dynamic optimization for the MIOR injection concentrations in the synthetic reservoir after waterflooding. As shown, the optimization algorithm pushed for the lower concentrations for *Bacillus* and glucose with higher concentration for sulfate injections due to their relative costs while maintaining efficient bio-surfactant generations for the optimal oil recovery. Fig. 2 shows the cumulative oil production in unit of percentage of original oil in place (%OOIP) for the pure waterflood, MIOR without optimization, and MIOR with optimization cases. As shown, MIOR without and with optimization can boost final cumulative oil production from 60 to 78 and 80 %OOIP, respectively. Furthermore, Fig. 3 shows the final NPV for the pure waterflood, MIOR without optimization, and MIOR with optimization cases. As shown, MIOR without and with optimization can result in +512% and +1154% NPV increases compared to pure waterflooding. These results are encouraging for the further development of MIOR for practical applications.

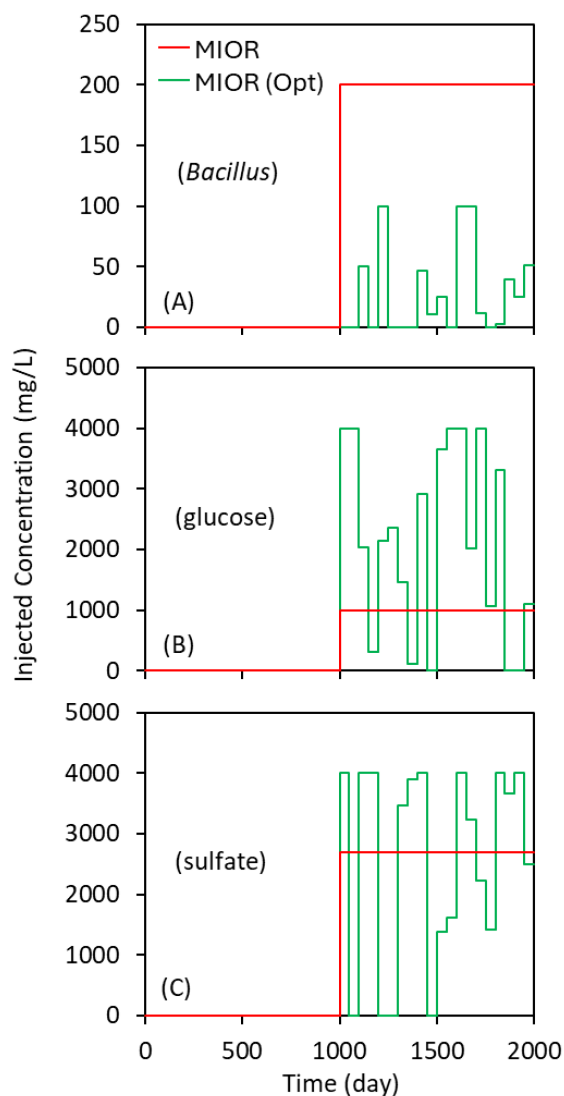


Figure 1—Concentrations as a function of time for the constant as well as optimized MIOR injections for (A) *Bacillus*, (B) glucose, and (C) sulfate.

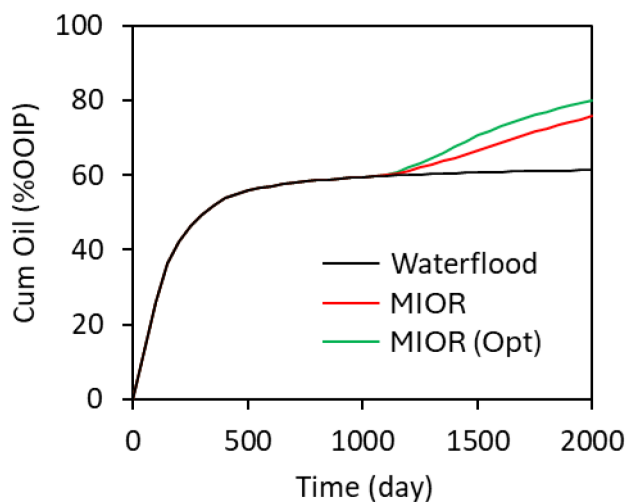


Figure 2—Cumulative oil production as a function of time in unit of percentage of original oil in place (%OOIP) for the pure waterflood, MIOR without optimization, and MIOR with optimization cases.

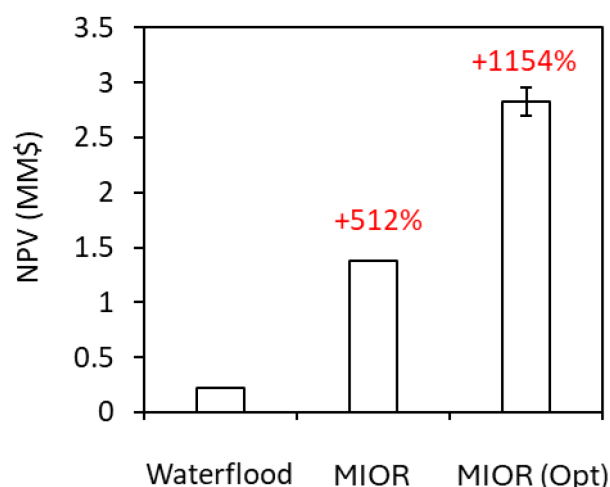


Figure 3—NPV for the pure waterflood, MIOR without optimization, and MIOR with optimization cases.

Finally, Fig. 4 shows the sensitivity studies with varying *Bacillus* costs (\$10/kg to \$50/kg), indigenous *Bacillus* concentrations in reservoir (0 to 10 mg/L), and oil prices (\$40/bbl to \$100/bbl). As shown, in all cases the MIOR with optimization can result in consistent +1000% NPV increases, which suggested the robustness of the present EnOpt algorithm and the high financial potential for the MIOR process.

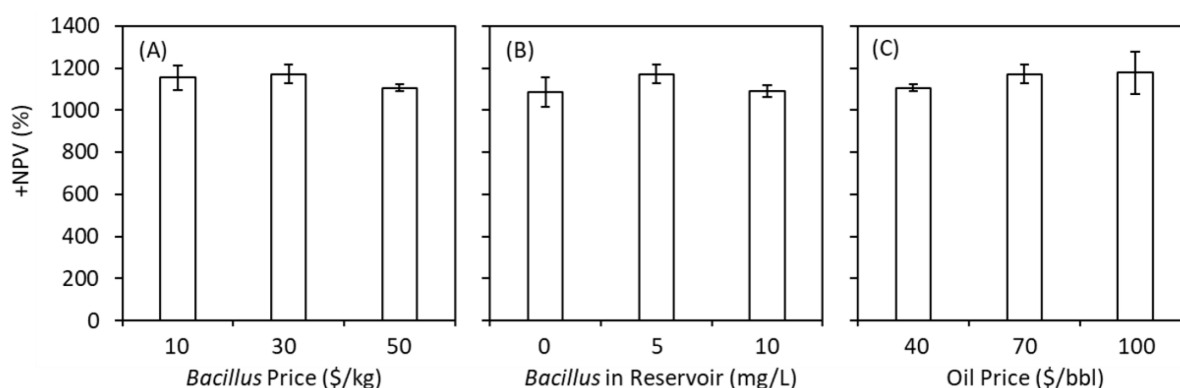


Figure 4—NPV results for the sensitivity studies with varying (A) *Bacillus* costs, (B) indigenous *Bacillus* concentrations in reservoir, and (C) oil prices.

Conclusions

The study highlights the significant potential of Microbial Improved Oil Recovery (MIOR) when optimized with advanced algorithms. Using the Ensemble Optimization (EnOpt) algorithm, MIOR achieved a 10-fold increase in net present value (NPV) and boosted oil recovery by 20% compared to waterflooding. Sensitivity studies confirmed the robustness of these financial benefits across various conditions. This novel application of EnOpt to MIOR demonstrates its effectiveness in enhancing oil recovery and financial returns, suggesting promising prospects for wider adoption and further development of MIOR technologies.

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